HVAC Remote Monitoring Project

Parts and Materials List | Portfolio | October 2026

This list records the parts selected for my prototype and the parts I want to add later. The temperature monitoring is working; the current sensing circuit is still being tested.

1. Prototype parts

Part	Quantity / status	What I use it for
ESP32 DevKitC-32UE	1; used	Reads the DS18B20 and sends live readings through USB.
DS18B20 waterproof temperature probes	2 sets of 5 ordered; 1 demonstrated	Measures temperature. The prototype currently shows live readings from one probe.
4.7 kΩ resistor	1 used; extras ordered	Pull-up for the DS18B20 data connection.
ADS1115 16-bit ADC	1; connected, current testing unfinished	Analog input converter for the current sensing circuit.
SCT013 current clamp, marked 100A/1V	1; testing unfinished	Intended to measure AC current; no verified current measurements yet.
10 kΩ resistors	2–3 specified	Parts of the current-sensor signal conditioning circuit.
10 µF capacitors	2 used in reported wiring	Parts of the current-sensor circuit.
Breadboard	1; used	Temporary wiring and testing.
ESP32 screw terminal breakout board	1; used	Makes ESP32 pins easier to connect to.
Jumper wires / Dupont connectors	Assorted; used	Prototype connections and testing.
USB cable	1; used	Power and live serial data to the computer.

2. Other base-version materials

Part	Quantity / status	What I use it for
Bud Industries DMB-4770 ABS DIN enclosure	1 specified; installation not shown	Housing for a later installed version.
H11AA1 transistor-output optocouplers	10 specified; not shown installed	Planned thermostat-call sensing.
2.2 kΩ resistors	10 specified; not shown installed	Associated planned circuit components.
Heat-shrink tubing	Assorted	Insulate finished wire connections.
M2.5 / M3 screws	Assorted	Mount parts in an enclosure.
Wire ferrules	Assorted; planned	Stronger terminations for a later build.

3. Future sensors and upgrades

Part	Quantity / status	What I use it for
SHT31 temperature / humidity breakout	1 planned	Add humidity measurements.

Sensirion SDP31 differential pressure sensor	1 planned	Explore pressure differences in HVAC airflow.
ADXL345 accelerometer	1 planned	Explore vibration measurements.
Mean Well HDR-15-5 5V power supply	1 planned	Power a standalone version.
TVS diode, input fuse, MOV	As needed; planned	Protection for a future installed version.
18/2 thermostat wire; 22/4 signal cable	As needed; planned	Longer permanent wiring runs.
DIN rail, cable glands, connectors	As needed; planned	Mounting and cable management.

4. What currently works

DS18B20 temperature sensor → ESP32 GPIO4 → USB connection → computer browser → live website temperature readings and graph.

The current clamp and ADS1115 wiring are part of the prototype, but the current readings are not yet working reliably. The photos and dashboard are evidence of the temperature measurements, not proof of completed HVAC fault diagnosis.

5. Source links for main parts

ESP32 DevKitC-32UE: https://www.digikey.com/en/products/detail/espressif-systems/ESP32-DEVKITC-32UE/12091813

DS18B20 probes: https://www.amazon.com/dp/B00M1PM55K

ADS1115 breakout: https://www.digikey.com/en/products/detail/adafruit-industries-llc/1085/5761229

Current clamp listing: https://www.amazon.com/dp/B083S6Y3DJ

10 kΩ resistors: https://www.digikey.com/en/products/detail/yageo/CFR-25JR-52-10K/11998

10 µF capacitors: https://www.digikey.com/en/products/detail/panasonic-industry/ECA-1HM100/245037

ESP32 terminal breakout: https://www.amazon.com/dp/B08LGQ2H72

Note: The original purchase list describes an SCT013-000 type, while the photographed clamp is labeled 100A/1V. The documented prototype uses the label visible on the actual clamp. Quantities ordered do not necessarily equal quantities installed.